Lightweight Voice Activity Detection Using Cepstral Features and Local Texture Encoding

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Abstract. Voice Activity Detection (VAD) is an important component in speech processing systems for distinguishing speech from non-speech signals. Although deep learning approaches have achieved strong performance, they often require large-scale datasets and high computational resources. This paper proposes a lightweight VAD framework based on handcrafted cepstral features and Local Binary Pattern (LBP) encoding. The proposed framework systematically evaluates different spectral representations, including MFCC and LFCC, with coefficient settings of 20, 40, and 80. LBP-based texture encoding is applied to the extracted feature maps to capture local structural variations within cepstral representations. Unlike conventional approaches that mainly rely on global spectral characteristics, the proposed framework explicitly models local neighborhood patterns for speech/non-speech discrimination. Experimental results on the MUSAN dataset demonstrate strong performance across different classifiers and feature settings. Among all evaluated configurations, XGBoost combined with LFCC-LBP achieved the best performance with an accuracy of 98.8% and an Equal Error Rate (EER) of 1.1%. The results further show that LFCC consistently outperformed MFCC, suggesting that linear-frequency representations are more suitable for LBP-based texture representations in VAD tasks. Compared with previous studies reporting approximately 89% accuracy on the same dataset, the proposed framework achieves substantial performance improvement while maintaining a lightweight and computationally efficient design. These findings demonstrate the effectiveness of combining cepstral features and local texture encoding for practical VAD applications.

Keywords: Voice Activity Detection, Handcrafted Features, MFCC, LFCC, LBP, Cepstral Features, Machine Learning

1 Introduction

Voice Activity Detection (VAD) is a fundamental component in many speech processing systems, including automatic speech recognition, speaker verification,

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speech enhancement, telecommunication systems, and voice assistants. Its primary objective is to accurately distinguish speech segments from non-speech regions such as silence, background noise, and music. Reliable VAD not only improves the performance of downstream speech applications but also reduces unnecessary computation by processing speech-active regions only.

Over the past decades, a wide range of VAD approaches has been proposed. Early signal-based methods mainly relied on handcrafted measurements such as short-time energy, zero-crossing rate, spectral centroid, and entropy measures. These techniques are computationally efficient and suitable for real-time systems; however, their performance often degrades significantly under low signal-to-noise ratio (SNR) or non-stationary noise conditions [1]. To improve robustness, statistical model-based approaches were introduced using likelihood ratio tests, adaptive noise estimation [2]. Although these methods outperform simple threshold-based techniques in moderate noise environments, their effectiveness strongly depends on distributional assumptions and accurate parameter estimation.

More recently, machine learning and deep learning approaches have significantly advanced VAD performance. Conventional machine learning methods using handcrafted spectral features combined with classifiers such as SVM, Random Forest, and ensemble learning models have demonstrated strong effectiveness for speech activity classification [3,4]. Among handcrafted representations, MFCC remains one of the most widely used speech features due to its perceptual relevance and compact representation capability [5]. LFCC preserves linearly spaced frequency information and has shown effectiveness in several speech-related tasks [6]. In parallel, deep learning architectures, including CNN-, RNN-, CRNN-, and Transformer-based models, have achieved state-of-the-art performance by automatically learning discriminative representations from spectrograms or raw waveforms [7,8]. However, these approaches generally require large-scale labeled datasets, expensive training procedures, and high computational resources, which may limit their applicability in lightweight or low-resource environments.

Despite the strong performance of existing methods, feature representation remains a critical challenge for lightweight VAD systems. Traditional handcrafted features such as MFCC and LFCC mainly focus on global spectral characteristics and may not fully exploit local structural variations in the time–frequency domain. However, local patterns such as harmonic continuity, transient energy changes, and texture differences between speech and non-speech regions may provide important discriminative cues for robust VAD. While deep learning models can automatically learn such local structures, they typically introduce significant computational complexity and reduce interpretability.

Motivated by these limitations, there is a need for a lightweight and interpretable framework that can enhance handcrafted spectral representations while explicitly modeling local structural information. In computer vision and audio analysis, texture descriptors have been widely used to capture discriminative local spatial patterns. Among them, Local Binary Pattern (LBP) is a simple yet effective operator that encodes local neighborhood variations with low computational complexity [10]. Previous studies have demonstrated that LBP-based texture representations can effectively characterize spectrogram-like patterns in audio signals. In particular, Costa et al. [11] successfully applied LBP textural features for music genre classification, showing that local texture patterns extracted from time–frequency representations provide strong discriminative information. Inspired by this observation, we investigate whether LBP can also enhance handcrafted speech spectral representations for lightweight Voice Activity Detection (VAD). Although LBP has shown effectiveness in several audio and image analysis tasks, its application to MFCC/LFCC-based speech activity classification remains relatively underexplored.

Therefore, this paper proposes a lightweight VAD framework that integrates handcrafted spectral features with LBP-based texture encoding. Specifically, MFCC and LFCC feature maps are transformed using LBP to capture discriminative local structural patterns in the time–frequency domain. The resulting histogram-based representations are subsequently classified using conventional machine learning models, including Random Forest, XGBoost, SVM, KNN and Logistic Regression. By combining efficient handcrafted spectral representations with local texture modeling, the proposed framework aims to improve speech/non-speech discrimination while maintaining low computational complexity.

The main contributions of this work are summarized as follows:

- We propose a lightweight VAD framework that integrates handcrafted spectral representations with LBP-based local texture encoding.
- We investigate the effectiveness of LBP for enhancing MFCC and LFCC representations in speech/non-speech classification.
- We provide a unified comparison between different cepstral representations under the same processing pipeline.
- We demonstrate that LBP-enhanced spectral features combined with conventional machine learning models achieve strong VAD performance on the MUSAN dataset.

This paper is organized as follows. Section 2 presents the proposed methodology. Section 3 describes the experimental setup. Section 4 presents and discusses the experimental results. Finally, Section 5 concludes the paper.

2 Methodology

2.1 Overview of the proposed framework

Figure 1 demonstrates the overall pipeline of the proposed voice activity detection (VAD) framework. The MUSAN dataset is first divided using a K-fold cross-validation strategy to construct training and testing subsets for in-domain evaluation.

For each training fold, the audio signals are preprocessed and transformed into two different time–frequency representations, including MFCC and LFCC.

Local Binary Pattern (LBP) encoding is then applied to these feature maps to capture discriminative local structural patterns. The resulting histogram-based feature vectors are subsequently used to train several machine learning classifiers, including XGBoost, Random Forest, SVM, KNN, and Logistic Regression.

Finally, the trained models are evaluated on both the MUSAN testing subset and the Aurora-2 dataset using standard evaluation metrics such as Accuracy, F1-score, and Equal Error Rate (EER).

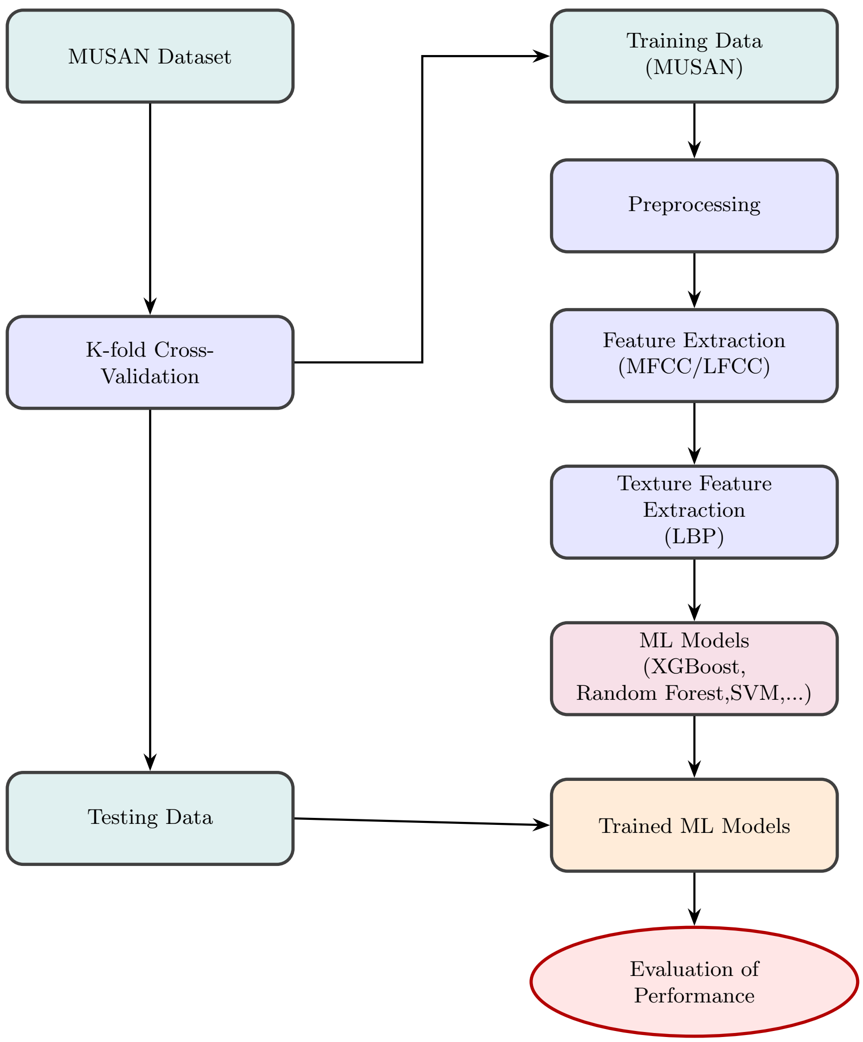

Fig. 1. Proposed framework for Voice Activity Detection

2.2 Data Preprocessing

Before feature extraction, two preprocessing steps are applied to the input audio signals, including silence removal and audio segmentation.

For silence removal, the audio signal is divided into short overlapping frames using a window length of 25 ms with a 15 ms overlap. The short-time energy of each frame is computed as

$$E = \sum_{n=0}^{N-1} x[n]^2$$

where $x[n]$ denotes the audio signal samples within a frame and N represents the frame length. Frames with low energy are removed to reduce silent regions and background interference.

To increase the number of training samples and ensure a consistent input duration, all recordings are segmented into fixed-length audio clips of 2 seconds. Each segment contains either speech or non-speech audio only.

2.3 Feature Extraction Methods

To represent the acoustic characteristics of speech signals, three handcrafted time–frequency representations are extracted from short audio frames, including Mel-Frequency Cepstral Coefficients (MFCC) and Linear Frequency Cepstral Coefficients (LFCC). These features capture complementary spectral and temporal information for speech/non-speech discrimination. In addition, Local Binary Pattern (LBP) encoding is applied to the extracted feature maps to model local structural patterns in the time–frequency domain.

Mel-Frequency Cepstral Coefficients (MFCC) MFCC is one of the most widely used feature representations in speech and audio processing. It models human auditory perception by mapping the linear frequency spectrum onto the Mel scale. The MFCC extraction process typically involves computing the Short-Time Fourier Transform (STFT), applying a Mel-scaled filter bank, taking the logarithm of filter-bank energies, and performing a Discrete Cosine Transform (DCT) to obtain compact cepstral coefficients that describe the spectral envelope of speech [5].

Linear Frequency Cepstral Coefficients (LFCC) LFCC is a cepstral representation similar to MFCC but uses linearly spaced filter banks instead of Mel-scaled filters. By preserving the linear frequency distribution, LFCC retains more detailed information in higher-frequency regions of the spectrum. This property makes LFCC suitable for tasks such as speaker verification and spoofing detection, where high-frequency artifacts may provide important discriminative information [6].

For feature extraction, both MFCC and LFCC were computed using a frame-based analysis with a window length of 25 ms and a 15 ms overlap. To investigate the effect of feature dimensionality, three different configurations were considered using 20, 40, and 80 cepstral coefficients.

Local Binary Pattern (LBP) LBP is a texture descriptor used to capture local spatial patterns in images. It operates by comparing the intensity value of a central pixel with its neighboring pixels and encoding the comparison results into a binary pattern that represents local texture information. By computing histograms of these binary patterns over an image or feature map, LBP effectively characterizes structural variations and has been widely used in pattern recognition tasks such as texture classification and face recognition [10].

In this study, the input audio waveform is first transformed into handcrafted spectral representations using MFCC and LFCC. Each audio segment is converted into a two-dimensional feature matrix, where rows correspond to temporal frames and columns correspond to cepstral coefficients, as illustrated in Fig. 2.

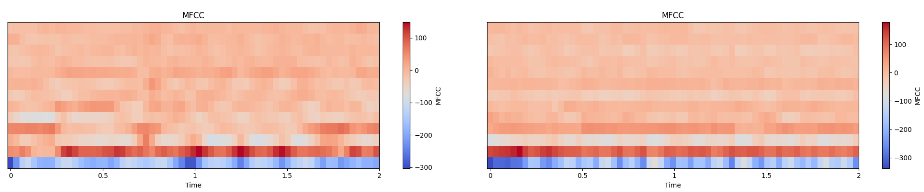

Fig. 2. Cepstral feature matrices (left: speech, right: non-speech)

LBP encoding is then applied to the extracted feature matrices to capture local structural variations in the time–frequency domain. In this work, the original LBP operator was configured with $P = 8$ sampling points and radius $R = 1$. The resulting binary patterns are summarized using histogram representations, producing a 256-dimensional feature vector that serves as the final input for classification, as shown in Fig. 3.

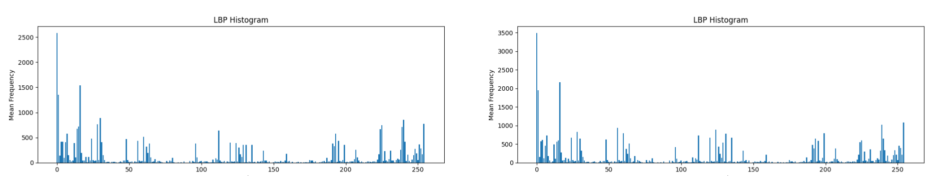

Fig. 3. LBP histogram features (left: speech, right: non-speech)

2.4 Classification Models

To evaluate the effectiveness of the proposed feature representations, five conventional machine learning classifiers are employed, including KNN, LR, SVM, RF, and XGBoost. These models represent different learning strategies, including distance-based, linear, margin-based, and ensemble methods.

K-Nearest Neighbors (KNN) KNN is a distance-based and instance-based learning algorithm that classifies samples according to the labels of their nearest neighbors in the feature space. Despite its simplicity, KNN can achieve strong performance when discriminative feature representations are available [12].

Logistic Regression (LR) LR is a linear statistical classifier that estimates the probability of a sample belonging to a particular class using the logistic sigmoid function. Due to its computational efficiency and interpretability, LR is widely used as a lightweight baseline for binary classification tasks [13].

Support Vector Machine (SVM) SVM is a supervised learning algorithm that determines an optimal decision boundary by maximizing the margin between classes. SVM is particularly effective for high-dimensional feature spaces and can handle non-linear classification problems through kernel functions [14].

Random Forest (RF) RF is an ensemble learning algorithm based on the bagging strategy, where multiple decision trees are trained using randomly sampled data and feature subsets. The final prediction is obtained through majority voting, which improves robustness and reduces overfitting [15].

Extreme Gradient Boosting (XGBoost) XGBoost is an optimized gradient boosting framework that constructs decision trees sequentially to correct errors from previous trees. By incorporating regularization and efficient optimization techniques, XGBoost achieves strong predictive performance and scalability [16].

3 Experimental Setup

The experimental pipeline consists of two main stages: dataset preparation, performance Evaluation.

3.1 Dataset

In this study, the MUSAN dataset is used to evaluate the proposed VAD framework [9]. MUSAN is a widely used audio corpus containing speech, music, and noise recordings, including approximately 60 hours of speech, 42 hours of music, and 7 hours of noise data.

For the VAD task, the speech subset is treated as the speech class, while the music and noise subsets are combined and considered as the non-speech class. After segmentation, the dataset contains 64,017 speech segments and 62,696 non-speech segments.

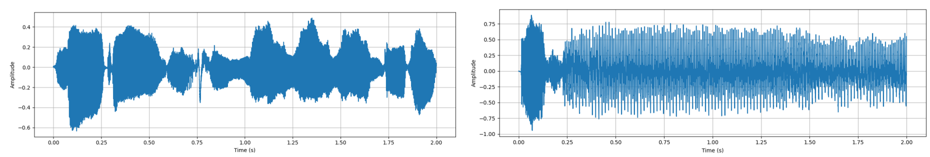

Fig. 4. Visualization of Speech's segment and Non-speech's segment in waveform (left: speech, right: non-speech)

3.2 Performance Evaluation

The performance of the proposed framework was evaluated using a 5-fold cross-validation protocol on the MUSAN dataset. The entire dataset was divided into five non-overlapping subsets. In each iteration, four subsets were used for training and the remaining subset was used for testing. The final performance was obtained by averaging the results across all folds.

To evaluate classification effectiveness, five standard metrics were employed, including Accuracy, F1-score, and Equal Error Rate (EER). Accuracy measures the overall proportion of correctly classified samples, while F1-score provides a balanced measure between Precision and Recall, which evaluate the reliability of positive predictions and the ability to correctly detect speech segments, respectively.

In addition, Equal Error Rate (EER) was used to evaluate the trade-off between false positive and false negative rates. EER corresponds to the operating point where the False Positive Rate (FPR) equals the False Negative Rate (FNR). A lower EER value indicates better discrimination capability between speech and non-speech classes. EER is defined as follows:

$$\mathrm{EER} = \mathrm{FPR}(\tau^*) \quad \text{where} \quad \tau^* = \arg\min_{\tau} |\mathrm{FPR}(\tau) - \mathrm{FNR}(\tau)| \tag{1}$$

4 Results and Discussion

This section presents the experimental results obtained on the MUSAN dataset using different acoustic feature representations and conventional machine learning models. Table 4 summarizes the results achieved by XGBoost, RF, SVM, KNN, and LR on the MUSAN dataset. Overall, most configurations achieved strong performance, with many settings obtaining accuracy above 95%, demonstrating the effectiveness of the proposed LBP-based framework for speech/non-speech discrimination. The relatively low EER values further indicate reliable class separation capability.

Table 1. Results of Machine Learning Models (%)

Model	Feature	numCoef	Accuracy	F1	EER
XGBoost	MFCC	20	97.1 ± 0.0006	97.3 ± 0.0014	2.8 ± 0.0006
	MFCC	40	97.3 ± 0.0008	97.7 ± 0.0008	2.5 ± 0.0007
	MFCC	80	96.8 ± 0.0011	96.5 ± 0.0007	3.1 ± 0.0009
	LFCC	20	98.0 ± 0.0008	98.0 ± 0.0015	1.9 ± 0.0008
	LFCC	40	98.1 ± 0.0005	98.0 ± 0.0004	1.7 ± 0.0006
	LFCC	80	**98.8 ± 0.0005**	**98.7 ± 0.0012**	**1.1 ± 0.0004**
RF	MFCC	20	97.0 ± 0.0007	97.3 ± 0.0013	2.9 ± 0.0007
	MFCC	40	97.3 ± 0.0004	97.4 ± 0.0016	2.6 ± 0.0005
	MFCC	80	96.7 ± 0.0006	96.3 ± 0.0006	3.3 ± 0.0004
	LFCC	20	97.4 ± 0.0011	97.6 ± 0.0011	2.5 ± 0.0010
	LFCC	40	97.3 ± 0.0008	97.2 ± 0.0005	2.6 ± 0.0007
	LFCC	80	97.9 ± 0.0009	97.7 ± 0.0010	1.9 ± 0.0008
SVM	MFCC	20	95.3 ± 0.0010	95.7 ± 0.0018	4.7 ± 0.0010
	MFCC	40	95.2 ± 0.0007	95.5 ± 0.0015	4.8 ± 0.0005
	MFCC	80	94.5 ± 0.0004	94.0 ± 0.0017	5.5 ± 0.0009
	LFCC	20	96.9 ± 0.0005	96.8 ± 0.0012	3.0 ± 0.0007
	LFCC	40	96.8 ± 0.0007	96.6 ± 0.0017	3.2 ± 0.0007
	LFCC	80	97.7 ± 0.0005	97.4 ± 0.0010	2.3 ± 0.0008
KNN	MFCC	20	96.8 ± 0.0010	96.8 ± 0.0022	3.1 ± 0.0010
	MFCC	40	97.3 ± 0.0012	97.2 ± 0.0018	2.8 ± 0.0011
	MFCC	80	96.5 ± 0.0007	96.2 ± 0.0007	3.5 ± 0.0009
	LFCC	20	93.7 ± 0.0013	93.1 ± 0.0019	6.3 ± 0.0011
	LFCC	40	93.4 ± 0.0006	94.8 ± 0.0012	6.5 ± 0.0013
	LFCC	80	96.5 ± 0.0004	97.0 ± 0.0005	3.5 ± 0.0004
LR	MFCC	20	95.3 ± 0.0009	95.7 ± 0.0017	4.7 ± 0.0010
	MFCC	40	95.2 ± 0.0007	95.4 ± 0.0013	4.8 ± 0.0006
	MFCC	80	94.5 ± 0.0006	94.1 ± 0.0017	5.5 ± 0.0007
	LFCC	20	96.9 ± 0.0007	96.8 ± 0.0016	3.1 ± 0.0007
	LFCC	40	96.8 ± 0.0007	96.7 ± 0.0020	3.2 ± 0.0007
	LFCC	80	97.7 ± 0.0010	97.4 ± 0.0014	2.3 ± 0.0007

Among all evaluated configurations, XGBoost achieved the best overall performance. Using LFCC with 80 coefficients, XGBoost obtained the highest accuracy of 98.8% and F1-score of 98.7%, while achieving the lowest EER of 1.1%. Random Forest also showed competitive performance, reaching 97.9% accuracy with LFCC-80. In general, LFCC features consistently produced better results than MFCC across most classifiers and coefficient settings. The best MFCC configurations achieved approximately 97.3% accuracy, whereas LFCC reached up to 98.8%. Increasing the number of coefficients from 20 to 40 or 80 generally improved LFCC performance, although the trend was not always consistent for MFCC-based features.

Figure 5 further illustrates the feature distribution after dimensionality reduction. The visualization shows that speech and non-speech samples form relatively distinguishable clusters with limited overlap in several regions of the feature space. This separation indicates that the proposed LFCC-LBP representation effectively captures discriminative local spectral patterns between the two classes, which contributes to the strong classification performance achieved by ensemble learning models such as XGBoost and RF.

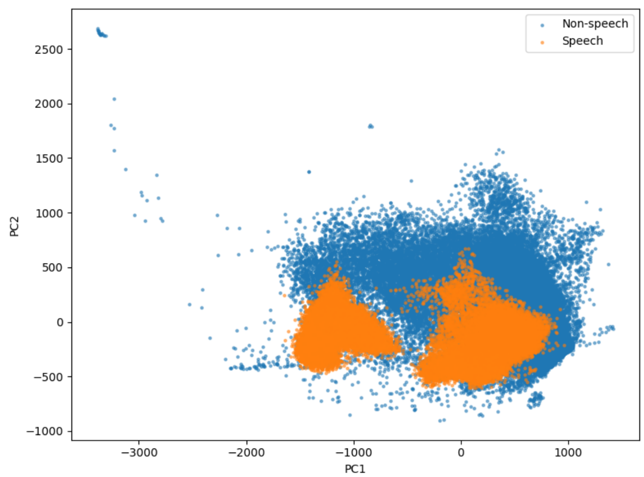

Fig. 5. LFCC 80 Coefficients Feature Distribution for Speech and Non-Speech

Interestingly, increasing the number of coefficients did not always improve performance, particularly for MFCC representations. Higher-dimensional feature vectors may introduce redundant or less informative spectral components, which can negatively affect conventional classifiers and increase sensitivity to noise. The small standard deviations across repeated runs indicate that the proposed framework provides stable and consistent performance across different experimental trials.

Experimental results suggest that feature representation plays a major role in VAD performance. One possible explanation for the superior performance of LFCC is that it preserves spectral resolution more uniformly across frequency bands due to its linear filter spacing. This characteristic allows local spectral transitions and texture-like structures to remain more distinguishable after feature extraction. Since LBP operates by encoding local neighborhood variations, richer local spectral patterns may provide more discriminative texture representations for VAD.

Tree-based ensemble methods, particularly XGBoost and RF, demonstrated stronger performance than SVM and LR. This suggests that the relationship between LBP-based texture descriptors and speech activity labels may be highly nonlinear. Ensemble learning methods are better suited to capture such complex decision boundaries while maintaining stable generalization performance across different feature settings.

4.1 Comparison with previous studies

In addition to evaluating different feature representations and classifiers, we further compared the proposed framework with several previous studies reported on the MUSAN dataset. The comparison in Fig. 6 demonstrates the effectiveness of the proposed approach for voice activity detection. The study by Dey et al. [4] reported approximately 89% accuracy using ensemble SVM with MFCC features and around 86% accuracy when combining neural networks with MFCC features. In contrast, the proposed framework achieved up to 98.8% accuracy using LFCC-80 with LBP and XGBoost.

The performance improvement may be attributed to the combination of discriminative spectral features and texture-based LBP encoding. By modeling local neighborhood patterns within spectral representations, the proposed framework is able to capture more distinctive speech and non-speech characteristics. In addition, tree-based classifiers such as XGBoost and RF are effective in handling nonlinear histogram-based feature spaces.

These results indicate that combining LFCC, LBP, and conventional machine learning classifiers provides a robust and computationally efficient framework for practical voice activity detection applications.

5 Conclusion

This paper presents a VAD framework based on handcrafted spectral features, LBP encoding, and conventional machine learning classifiers. The proposed framework comprehensively evaluates various spectral representations, specifically MFCC and LFCC, with coefficient settings of 20, 40, and 80. These features are transformed using LBP-based texture encoding prior to classification using RF, XGBoost, SVM, KNN, and LR. Experimental results on the MUSAN dataset demonstrate robust performance across different classifiers and feature settings. Notably, XGBoost combined with LFCC-LBP achieved the highest accuracy of 98.8% and the lowest EER of 1.1%. The findings reveal that LFCC consistently outperforms MFCC, suggesting that linear-frequency representations are better suited for LBP-based texture representations in VAD tasks. Compared with previous studies reporting approximately 89% accuracy on the MUSAN dataset, the proposed framework achieves a substantial performance improvement while maintaining a lightweight and computationally efficient design for practical VAD applications. Future research will focus on evaluating the proposed framework on additional VAD datasets and conducting cross-dataset experiments to assess

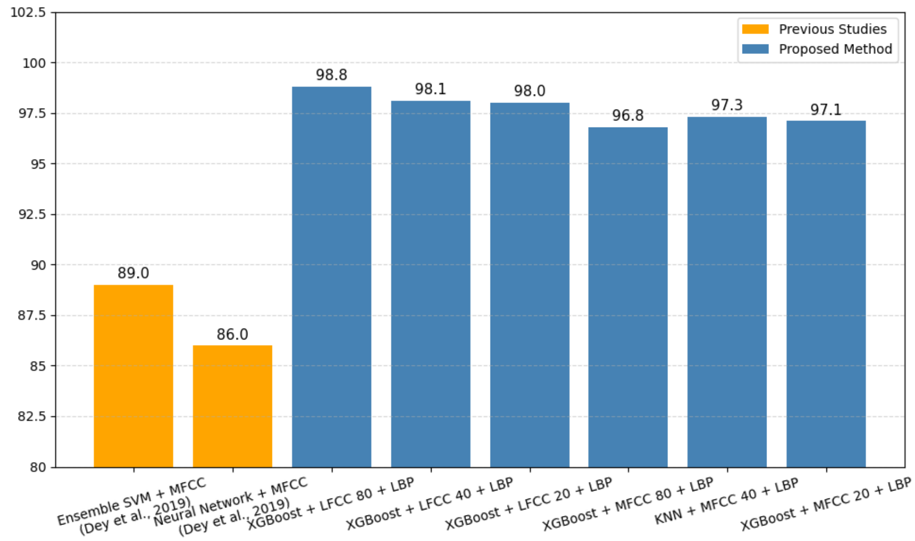

Fig. 6. Comparision of Accuracy on Musan Dataset

generalization capability further. More advanced texture descriptors, such as Completed Local Binary Pattern (CLBP), will also be investigated to improve the representation of local spectral structures under challenging real-world noise conditions.

References

1. Ramírez, J., Segura, J.C., Benítez, C., de la Torre, A., Rubio, A.: Efficient voice activity detection algorithms using long-term speech information. Speech Communication **42**(3–4), 271–287 (2004)
2. Sohn, J., Kim, N.S., Sung, W.: A statistical model-based voice activity detection. IEEE Signal Processing Letters **6**(1), 1–3 (1999)
3. Kinnunen, T., et al.: Voice activity detection using MFCC features and SVM. In: Proceedings of SPECOM 2007, pp. 556–570 (2007)
4. Dey, J., Hossain, M.S.B., Haque, M.A.: An ensemble SVM-based approach for voice activity detection. arXiv preprint arXiv:1902.01544 (2019)
5. Davis, S.B., Mermelstein, P.: Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences. IEEE Trans. Acoust. Speech Signal Process. **28**(4), 357–366 (1980)
6. Homayounpour, M.M., Chollet, G.: A comparison of some relevant parametric representations for speaker verification. In: ESCA Workshop on Automatic Speaker Recognition, Identification and Verification, pp. 185–188 (1994)
7. Hughes, T., Mierle, K.: Recurrent neural networks for voice activity detection. Proceedings of ICASSP (2013)
8. Jia, Y., et al.: MarbleNet: Deep 1D time-channel separable convolutional neural network for voice activity detection. arXiv preprint arXiv:2010.13886 (2021)

9. Snyder, D., Chen, G., Povey, D.: MUSAN: A music, speech, and noise corpus. arXiv preprint arXiv:1510.08484 (2015)
10. Ojala, T., Pietikäinen, M., Harwood, D.: A comparative study of texture measures with classification based on featured distributions. Pattern Recognition **29**(1), 51–59 (1996)
11. Costa, Y.M.G., Oliveira, L.S., Koerich, A.L., Gouyon, F., Martins, J.G.: Music genre classification using LBP textural features. Signal Processing **92**(11), 2723–2737 (2012)
12. Cover, T., Hart, P.: Nearest neighbor pattern classification. IEEE Transactions on Information Theory **13**(1), 21–27 (1967)
13. Hosmer, D.W., Lemeshow, S., Sturdivant, R.X.: Applied Logistic Regression. John Wiley & Sons (2013)
14. Cortes, C., Vapnik, V.: Support-vector networks. Machine Learning **20**(3), 273–297 (1995)
15. Breiman, L.: Random forests. Machine Learning **45**(1), 5–32 (2001)
16. Chen, T., Guestrin, C.: XGBoost: A scalable tree boosting system. In: Proc. 22nd ACM SIGKDD, pp. 785–794 (2016)